	$\mathcal{K}^{\#1}_{0^+}$	$\mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi}$
$\mathcal{K}^{\#1}_{0^+}\dagger$	$-\frac{3 Y_1 k^2}{2}$	0
$\mathcal{K}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	$3 k^2 \overset{(4)}{\kappa}_{15} + 3 \overset{(2)}{\kappa}_{1}$

	$\mathcal{K}^{\#1}_{1^+\,\alpha\beta}$	$\mathcal{K}^{\#2}_{1^+\,\alpha\beta}$	$\mathcal{K}^{\#1}_{1^-\,\alpha}$	$\mathcal{K}^{\#2}_{1^-\,\alpha}$
$\mathcal{K}^{\#1}_{1^+}\dagger^{\alpha\beta}$	$Y_2 k^2 + 2 \overset{(2)}{\kappa}_{1}$	$\frac{Y_3 k^2 + 4 \overset{(2)}{\kappa}_{1}}{2\sqrt{2}}$	0	0
$\mathcal{K}^{\#2}_{1^+}\dagger^{\alpha\beta}$	$\frac{Y_3 k^2 + 4 \overset{(2)}{\kappa}_{1}}{2\sqrt{2}}$	$\frac{1}{2}(Y_4 k^2 + 2 \overset{(2)}{\kappa}_{1})$	0	0
$\mathcal{K}^{\#1}_{1^-}\dagger^{\alpha}$	0	0	$-k^2 \overset{(4)}{\kappa}_{2}$	$-\sqrt{2}\, k^2 \overset{(4)}{\kappa}_{2}$
$\mathcal{K}^{\#2}_{1^-}\dagger^{\alpha}$	0	0	$-\sqrt{2}\, k^2 \overset{(4)}{\kappa}_{2}$	$-2 k^2 \overset{(4)}{\kappa}_{2}$

	$\mathcal{K}^{\#1}_{2^+\,\alpha\beta}$	$\mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi}$
$\mathcal{K}^{\#1}_{2^+}\dagger^{\alpha\beta}$	0	0
$\mathcal{K}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{J}^{\#1}_{0^+}$	$\mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi}$
$\mathcal{J}^{\#1}_{0^+}\dagger$	$\frac{1}{\mathrm{Det}(0^+)}$	0
$\mathcal{J}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	$\frac{1}{3(k^2 \overset{(4)}{\kappa}_{15} + \overset{(2)}{\kappa}_{1})}$

	$\mathcal{J}^{\#1}_{1^+\,\alpha\beta}$	$\mathcal{J}^{\#2}_{1^+\,\alpha\beta}$	$\mathcal{J}^{\#1}_{1^-\,\alpha}$	$\mathcal{J}^{\#2}_{1^-\,\alpha}$
$\mathcal{J}^{\#1}_{1^+}\dagger^{\alpha\beta}$	$\frac{Y_4 k^2 + 2 \overset{(2)}{\kappa}_{1}}{2\,\mathrm{Det}(1^+)}$	$\frac{-Y_3 k^2 - 4 \overset{(2)}{\kappa}_{1}}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#2}_{1^+}\dagger^{\alpha\beta}$	$\frac{-Y_3 k^2 - 4 \overset{(2)}{\kappa}_{1}}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$\frac{Y_2 k^2 + 2 \overset{(2)}{\kappa}_{1}}{\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#1}_{1^-}\dagger^{\alpha}$	0	0	$-\frac{1}{9 k^2 \overset{(4)}{\kappa}_{2}}$	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_{2}}$
$\mathcal{J}^{\#2}_{1^-}\dagger^{\alpha}$	0	0	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_{2}}$	$-\frac{2}{9 k^2 \overset{(4)}{\kappa}_{2}}$

	$\mathcal{J}^{\#1}_{2^+\,\alpha\beta}$	$\mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi}$
$\mathcal{J}^{\#1}_{2^+}\dagger^{\alpha\beta}$	0	0
$\mathcal{J}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0

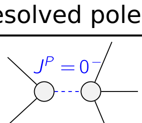
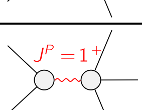
Abbreviations used in matrices

$$Y_1 == \overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2} \,\&\&\, Y_2 == 2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3} \,\&\&\, Y_3 == 4 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{4} \,\&\&\, Y_4 == 2 \overset{(4)}{\kappa}_{15} - 9 \overset{(4)}{\kappa}_{2} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4} \,\&\&$$
$$\mathrm{Det}(0^+) == -\frac{3}{2} k^2 (\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2}) \,\&\&\, \mathrm{Det}(1^+) == -\frac{1}{8} k^4 (72 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{2} - 36 \overset{(4)}{\kappa}_{2} \overset{(4)}{\kappa}_{3} + (2 \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2) - 9 k^2 \overset{(4)}{\kappa}_{2} \overset{(2)}{\kappa}_{1}$$

Lagrangian

$$\overset{(2)}{\kappa}_{1} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{\kappa}_{1} \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_{1} \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{2} \partial_\chi \mathcal{K}^{\delta}{}_{\beta\delta} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} +$$
$$\overset{(4)}{\kappa}_{4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} - \overset{(4)}{\kappa}_{3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - 6 \overset{(4)}{\kappa}_{2} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} \partial_\delta \mathcal{K}_{\chi\beta}{}^{\delta} - \frac{9}{2} \overset{(4)}{\kappa}_{2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} +$$
$$\frac{1}{2} \overset{(4)}{\kappa}_{3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2} \overset{(4)}{\kappa}_{4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	**# constraint(s)**	**Covariant form**
$\mathcal{J}^{\#1\,\alpha}_{1^-} == \mathcal{J}^{\#2\,\alpha}_{1^-}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} == 3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	13	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{\overset{(2)}{\kappa}_{1}}{\overset{(4)}{\kappa}_{15}}$	$-\frac{1}{3 \overset{(4)}{\kappa}_{15}}$
	3	$-\frac{72 \overset{(4)}{\kappa}_{2} \overset{(2)}{\kappa}_{1}}{72 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{2} - 36 \overset{(4)}{\kappa}_{2} \overset{(4)}{\kappa}_{3} + (2 \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2}$	$\frac{108 \overset{(4)}{\kappa}_{2}{}^2 - 12 \overset{(4)}{\kappa}_{2} (2 \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}) + (2 \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2}{3 \overset{(4)}{\kappa}_{2} (72 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{2} - 36 \overset{(4)}{\kappa}_{2} \overset{(4)}{\kappa}_{3} + (2 \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2)}$
Resolved unitarity condition(s):	(Demonstrably impossible)		